

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Scenario-Based
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
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Batch	AIML 2024	Date:	7/8/2026

Code of Conduct

I M.MANASA(192425189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: m.manasa

RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

9. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

(a) Interpret the scenario and explain why edges are broken.

(b) Identify key issues affecting boundary continuity.

(c) Apply a suitable method to improve edge connectivity.

(d) Justify your approach logically.

(e) Present your explanation clearly.

(a) Interpret the scenario and explain why edges are broken.

After applying an edge detection algorithm (such as Sobel, Prewitt, or Canny), the detected object boundaries appear discontinuous or broken instead of forming a complete outline.

This happens because:

- Noise in the image creates weak edge responses.
- Improper threshold values remove weak but important edge pixels.
- Low image contrast makes some boundaries difficult to detect.
- Gaps occur due to uneven illumination or texture variations.

Interpretation:

The edge detection process has identified only parts of the object's boundary, resulting in incomplete edges that make object recognition and segmentation difficult.

(b) Identify key issues affecting boundary continuity.

The main issues are:

1. Image Noise

- Random noise causes false or missing edges.

2. High Threshold Value

- Weak edges are removed during thresholding.

3. Low Image Contrast

- Small intensity differences reduce edge strength.

4. Incomplete Gradient Detection

- Some boundary pixels have weak gradients and are not detected.

5. Poor Image Quality

- Blurring and uneven lighting lead to discontinuous edges.

(c) Apply a suitable method to improve edge connectivity.

A suitable method is **Morphological Closing**. **Step 1:**

Broken Edge

1 1 0 1 1

Here, **0** represents a gap.

Step 2: Apply Morphological Closing

Closing = **Dilation** → **Erosion**

- **Dilation** fills the small gap.
- **Erosion** restores the original edge thickness while keeping the connection.

Output

1 1 1 1 1

The broken edge becomes a continuous boundary. Other

suitable methods include:

- Morphological Closing
- Edge Linking
- Canny Edge Detection with Hysteresis Thresholding

(d) Justify your approach logically.

Morphological Closing is the most suitable method because:

- It fills small gaps between disconnected edge pixels.

- It preserves the overall shape of the object.
- It removes minor discontinuities caused by noise.
- It improves boundary continuity, making segmentation and object recognition more accurate.

Therefore, morphological closing provides a reliable and effective solution for broken edges.

(e) Present your explanation clearly.

Conclusion:

Broken edges occur due to noise, improper threshold selection, low contrast, and weak gradient values. These discontinuities reduce the accuracy of image segmentation. Applying Morphological Closing (Dilation followed by Erosion) reconnects the broken boundaries, resulting in continuous and accurate object edges. This method enhances image quality and improves the performance of subsequent image processing tasks.